BCA PROJECT | DATA ANALYTICS, STATISTICAL ANALYSIS & MACHINE LEARNING

Airbnb India Market Intelligence

Pricing Analysis, Market Segmentation & Price Prediction



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Project Overview

What is Airbnb?

An online marketplace where hosts rent homes, apartments and rooms to travellers for short stays. Each host sets their own nightly price.

What this project analyses

102,599 Airbnb-style listings from five Indian cities: prices, locations, room types, booking rules, reviews and availability.

Final outcome

Pricing insights, statistical evidence, four market segments, a tested price-prediction model and two working Streamlit apps.

THE PROJECT IN NUMBERS

102,599
listings analysed

5 | 60
cities | neighbourhoods

₹2,750
median nightly price

₹954
average prediction error (test set)

Analytics explains the market; the model estimates a fair starting price for a listing.

Problem Statement

THE BUSINESS PROBLEM WE STARTED WITH

Prices differ by city

A night in Mumbai and a night in Jaipur are priced very differently.

Property type matters

Room type and listing features shape what guests will pay.

Hosts guess prices

New hosts struggle to set a competitive, fair price.

Raw data is not insight

Listing data must be cleaned and analysed before decisions.

WHAT THE DATA ACTUALLY ESTABLISHED

Room type is the biggest driver

Entire homes ≈ 2× private rooms (₹3,850 vs ₹1,900)

Location matters

Mumbai ₹3,450 vs Jaipur ₹2,200 median; neighbourhoods up to 1.56× apart

Ratings did not raise prices

Median ₹2,750 for every rating from 1 to 5

Prediction is a starting point

Model explains ~49% of price variation (R^2 0.486)

Project Objectives

1 COMPLETED

Analyse listing data

Clean and understand 102,599 listings

2 COMPLETED

Identify pricing patterns

Distribution, skew, typical price

3 COMPLETED

Compare cities & room types

City, neighbourhood and room-type gaps

4 COMPLETED

Test hypotheses

Seven statistical tests with effect sizes

5 COMPLETED

Predict nightly price

Four ML models vs a baseline

6 COMPLETED

Segment the market

K-Means clustering with PCA view

7 COMPLETED

Build dashboard & app

Two separate Streamlit applications

7 / 7

objectives completed and verified in the executed notebook

Dataset Description

102,599	25	5	60
rows (listings)	columns	cities	neighbourhoods

Target variable

price_inr_per_night: nightly price in ₹ (₹400 – ₹23,500)

Location

city, neighbourhood, latitude, longitude

Property & booking

room_type, construction_year, minimum_nights, instant_bookable, cancellation_policy, availability_365

Host & reviews

host_identity_verified, calculated_host_listings_count, number_of_reviews, reviews_per_month, review_rate_number

SAMPLE ROWS (FIRST 5 LISTINGS, SELECTED COLUMNS)

City	Neighbourhood	Room type	Min. nights	Reviews	Price (₹)
Hyderabad	Charminar	Private room	10	9	950
Bengaluru	HSR Layout	Entire home/apt	30	45	4,550
Mumbai	Chembur	Private room	3	0	1,500
Hyderabad	Banjara Hills	Entire home/apt	30	270	5,000
Bengaluru	MG Road	Entire home/apt	10	9	5,900

About the source

File supplied for the project (airbnb_india_listings.csv); original publisher not documented. Near-perfect category splits (e.g. 33.5 / 33.2 / 33.2% cancellation policies) suggest it may be synthetic, so results describe this dataset.

Technology Stack

Python

Core language for every step of the project

Pandas & NumPy

Load, clean, reshape and summarise the data

Matplotlib & Seaborn

Charts for EDA, models and clustering

SciPy

Hypothesis tests: t-test, ANOVA, chi-square

Scikit-learn

Pipelines, models, CV, K-Means and PCA

XGBoost

Gradient-boosted trees: the final model

Google Colab

Free cloud notebook to run the analysis

Streamlit

Turns Python into the two web apps

Also used: Plotly (interactive charts) and Joblib (saving the model bundle).

Project Workflow

1
Dataset
102,599 listings · 25 columns

2
Data Understanding
Types, ranges, quality checks

3
Data Cleaning
Logged rules, no rows lost

4
Feature Engineering
Analysis features + encoding

5
EDA
Price, city, room type, reviews

6
Statistical Analysis
7 tests with effect sizes

7
Machine Learning
4 models + baseline, CV, tuning

8
Clustering
K-Means, 4 segments, PCA

9
Dashboard & Prediction
Two tested Streamlit apps

10
Insights
Host & platform actions

Data Cleaning and Preprocessing

Missing values

Logical fill: 0 reviews ⇒ 0 reviews/month; rest imputed in the ML pipeline

Duplicates

Checked whole rows and listing IDs: none found

Invalid values

Rules for price, rating, nights, coordinates, dates

Outliers (IQR rule)

Prices above ₹7,550 flagged, kept as real premium listings

Type conversion

Numbers to numeric, review dates to datetime

Feature preparation

8 analysis columns; 25 ⇒ 34 columns

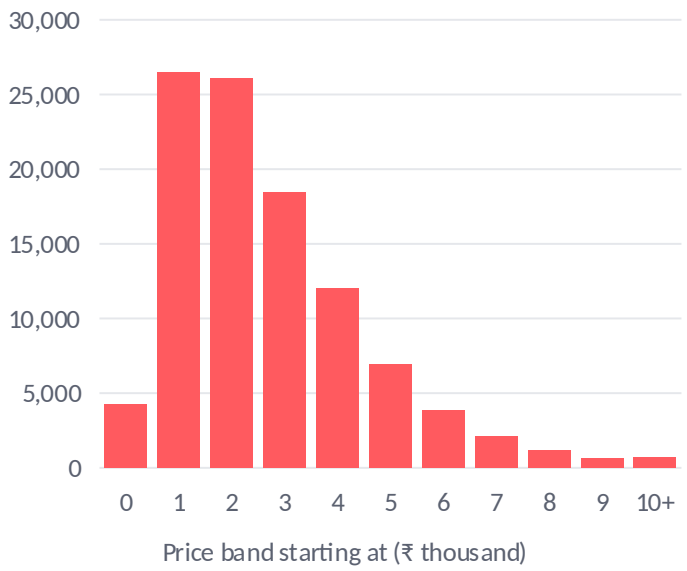
BEFORE AND AFTER CLEANING (ACTUAL COUNTS)

Check	Before	After
Rows (listings)	102,599	102,599
Duplicate rows / IDs	0	0
Invalid prices, ratings, coordinates	0	0
Review dates in the future (up to 2058)	5	**0 (set missing)**
reviews_per_month missing	15,879	**22**
Ratings missing	326	326 (imputed in pipeline)
High-price outliers flagged	–	3,168 (3.09%), kept

No rows deleted. Remaining gaps are filled inside the ML pipeline using training data only, which prevents data leakage.

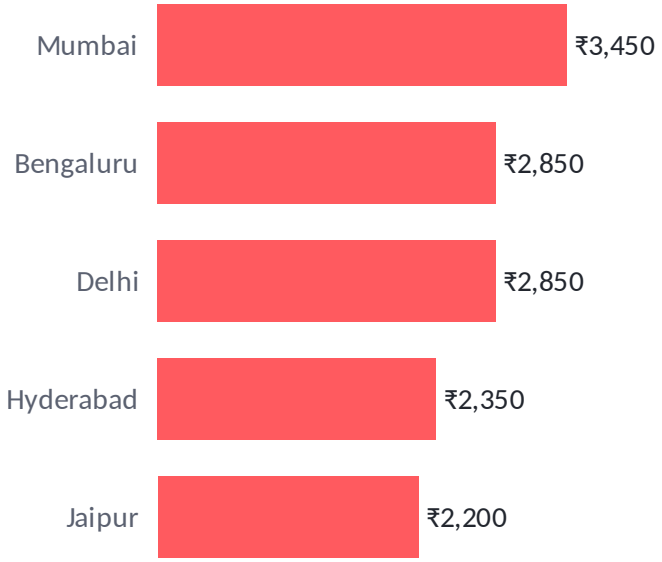
EDA: Price, City and Room Type

Price distribution (number of listings)



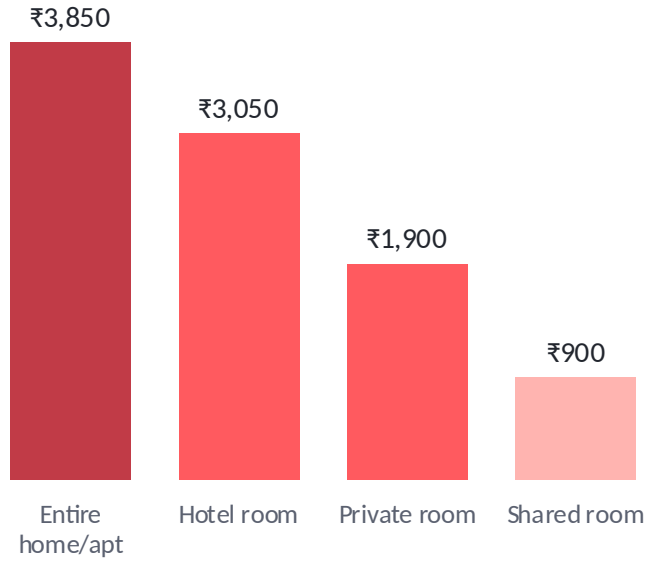
Right-skewed: most listings cost ₹1,000–4,000; median ₹2,750 is the typical price (mean ₹3,173).

Median price by city (₹ per night)



Mumbai highest (₹3,450), Jaipur lowest (₹2,200): about 57% apart.

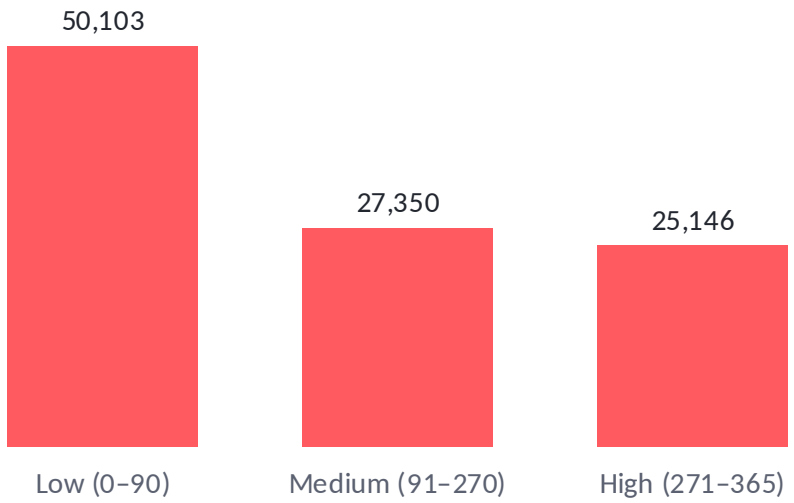
Median price by room type (₹ per night)



Entire home ≈ 2× private room (₹3,850 vs ₹1,900), and the ratio holds in every city.

EDA: Availability and Correlation

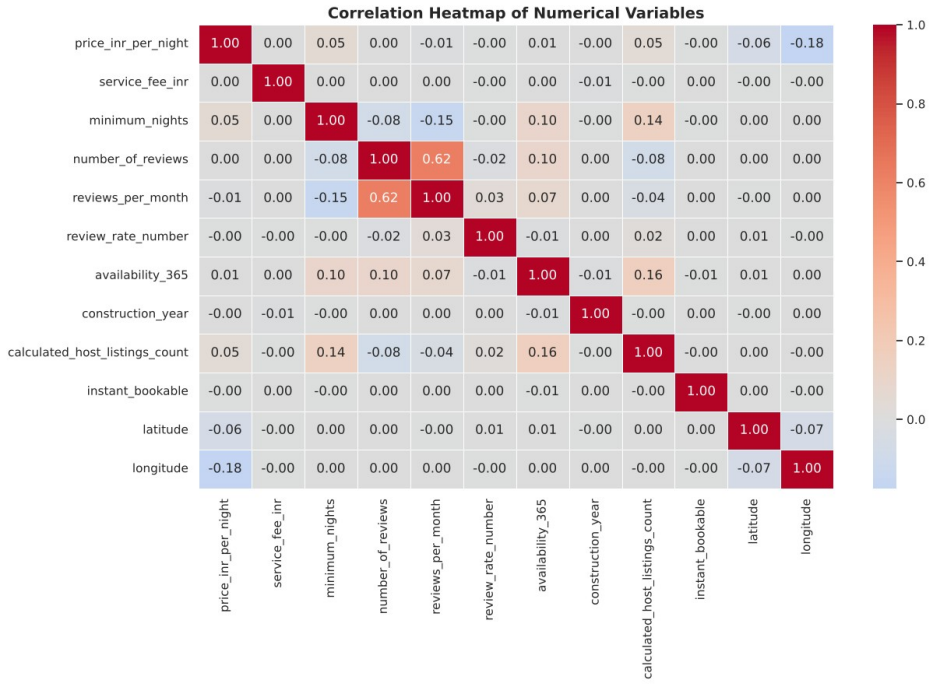
Listings by availability (days open per year)



Median price by group: ₹2,700 | ₹2,900 | ₹2,700

Availability barely changes price. Half of all listings are open 90 days or fewer, a first sign of dormant listings (confirmed by clustering).

Correlation heatmap (actual notebook output)



All |r| < 0.06 except longitude (−0.175), which reflects Mumbai being the westernmost and priciest city. Price is driven by categories (room type, location), not numbers like reviews.

Statistical Analysis

Test	Question	Result	Meaning
Shapiro-Wilk + Q-Q plot	Is price normally distributed?	W = 0.891, p ≈ 1e-50	No, so rank-based tests are also used
Pearson & Spearman	Is price linked to reviews or rating?	ρ = 0.019 (reviews); rating p ≈ 0.79	No meaningful link
Welch's t-test + Cohen's d	Entire home vs private room?	t = 232.2, d = 1.41	Large gap (mean ₹4,223 vs ₹2,066)
One-way ANOVA (η^2)	Do the 4 room types differ?	F = 18,395, η^2 = 0.350	Room type explains ~35% of variation
Kruskal-Wallis	Room types / cities differ (ranks)?	H = 46,994 / 7,167	Both differ; city effect smaller (ε^2 = 0.07)
Chi-square + Cramér's V	Is room-type mix linked to city?	p = 0.624, V = 0.006	Not linked: same mix in every city
95% confidence interval	Where is the true mean price?	₹3,162 – ₹3,185	Very precise, thanks to 1 lakh listings

What is a p-value?

The chance of seeing a difference this large if there were really no difference. p < 0.05 means 'statistically significant'. With 1 lakh listings even tiny effects become significant, so effect sizes are reported too.

Key finding

Room type has a large, real effect (Cohen's d 1.41, η^2 0.35). Reviews are 'significant' but negligible (ρ 0.019): significance alone does not mean importance.

Feature Engineering

Original columns

- number_of_reviews
- last_review (text date)
- review_rate_number
- availability_365
- host_id
- listing_name
- city, neighbourhood, room_type

Engineered for analysis

- has_reviews (0/1)
- log_number_of_reviews
- last_review_year, _month
- review_rate_missing (0/1)
- availability_category (Low/Med/High)
- host_category
- listing_name_length
- price_outlier_iqr (analysis only)

Prepared for the model

- Median / most-frequent imputation
- One-hot encoding: 13 inputs ⇒ 83 columns
- Standard scaling of numbers
- neighbourhood added as a feature
- latitude / longitude dropped (CV RMSE +₹0.4 only)
- All fitted on training data only

Why it helps: readable categories make EDA clearer; encoding and scaling let algorithms use text categories; leakage-safe rules keep test scores honest (the outlier flag is never a model input).

Machine Learning Approach

PROBLEM	TARGET	INPUTS	SPLIT
Supervised regression	price_inr_per_night (₹)	13 features (5 categorical, 8 numeric)	80/20 ⇒ 82,079 train | 20,520 test

Baseline

Predicts the median price for every listing: the bar to beat

Linear Regression

Price = weighted sum of features; simple and explainable

Decision Tree

A flowchart of yes/no questions that ends in a price

Random Forest

Averages 100 trees trained on random samples

XGBoost

Builds trees one after another, each fixing earlier errors

SHARED, LEAKAGE-SAFE PIPELINE (FITTED ON TRAINING DATA ONLY)

Listing features → Fill missing values → Scale numbers + one-hot categories → Model → Predicted ₹ price

Model Comparison

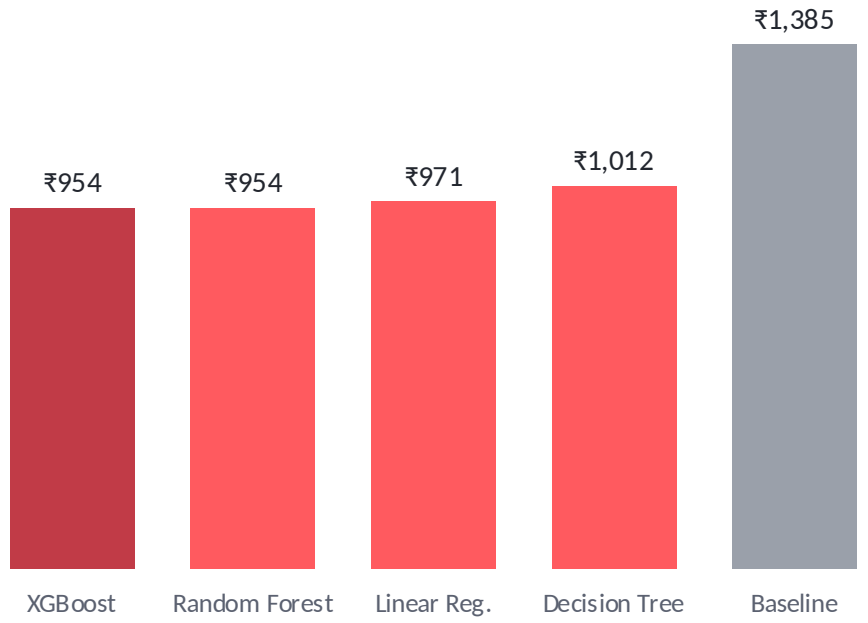
Model	Test MAE	Test RMSE	Test R^2	CV RMSE
XGBoost (selected)	**₹954**	**₹1,341**	**0.486**	**₹1,367**
Random Forest	₹954	₹1,341	0.486	₹1,368
Linear Regression	₹971	₹1,355	0.475	₹1,379
Decision Tree	₹1,012	₹1,434	0.412	₹1,461
Baseline (median)	₹1,385	₹1,916	−0.049	₹1,931

Test = 20,520 unseen listings. CV = 3-fold cross-validation on training data. Tuned versions changed CV RMSE by less than ₹2.

Why XGBoost was selected

Lowest cross-validation RMSE (₹1,367). Ties Random Forest on the test set but trains ~10× faster (6 s vs 56 s). Chosen by CV, so the test score stays an honest estimate.

Test MAE by model (₹, lower is better)



About 31% lower error than the baseline for the best models.

Final Model Evaluation

₹954

MAE

Average miss per listing (baseline ₹1,385)

₹1,341

RMSE

Like MAE, but big misses count more

0.486

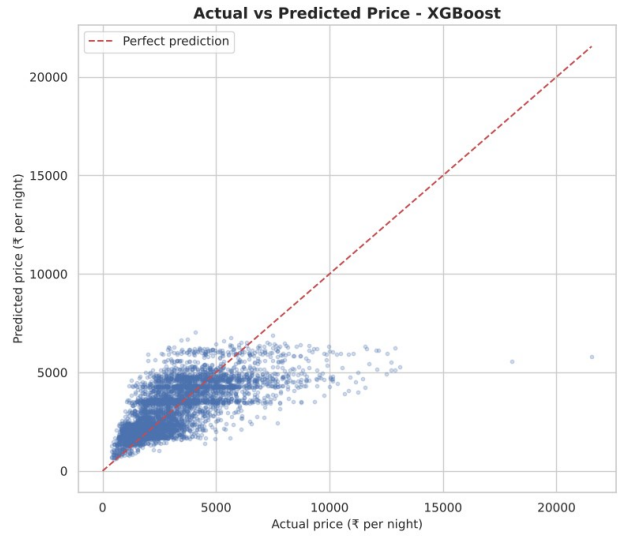
R^2

Explains ~49% of price differences

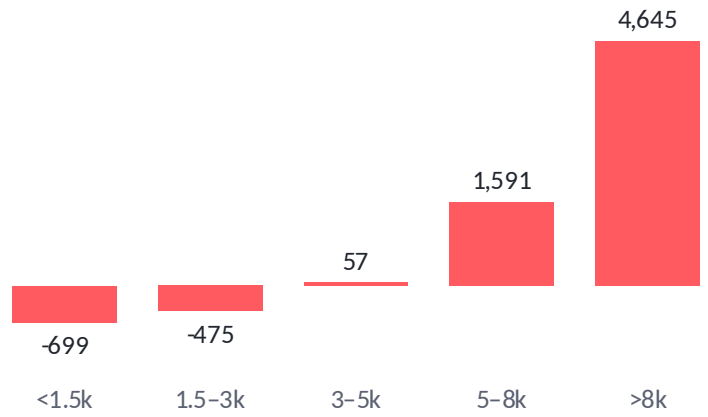
How it was evaluated

- 80/20 split, stratified by city
- 3-fold cross-validation
- Randomized search tuning
- Test set used only once

Actual vs predicted (test sample)



Average residual by actual price (₹)



Positive = predicted too low

Limitations

- Over-prices budget listings, under-prices luxury (≈ ₹4,600 low above ₹8,000)
- Half of price variation unexplained (no amenities, size, season)
- Tested only on unseen listings from this dataset

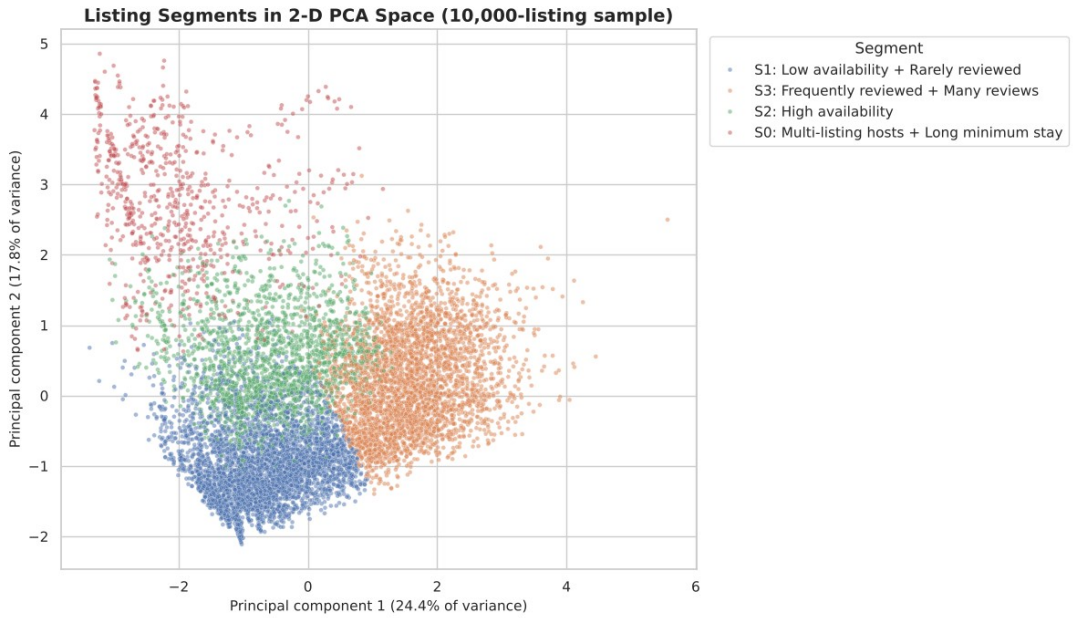
K-Means Clustering and Segmentation

ALGORITHM
K-Means on 8 standardised features

PRICE INCLUDED?
Yes: price-and-activity segments

CLUSTERS (K)
4 chosen; best silhouette K = 3

Segments in 2-D PCA space (actual notebook output)



Silhouette: K=3 → 0.192 (best), K=4 → 0.172. Kept 4 to separate dormant from open-but-unbooked listings.

32.5% ₹2,750 median

Active performers

Median 51 reviews, short stays

Action: Protect reviews, keep calendar open

39.2% ₹2,700 median

Dormant listings

Median 0 days available

Action: Reactivate or close the calendar

21.3% ₹2,650 median

Open but under-booked

313 days open, only 4 reviews

Action: Revisit price, photos, description

7.0% ₹3,200 median

Professional long-stay

30-night minimum, 50 listings/host

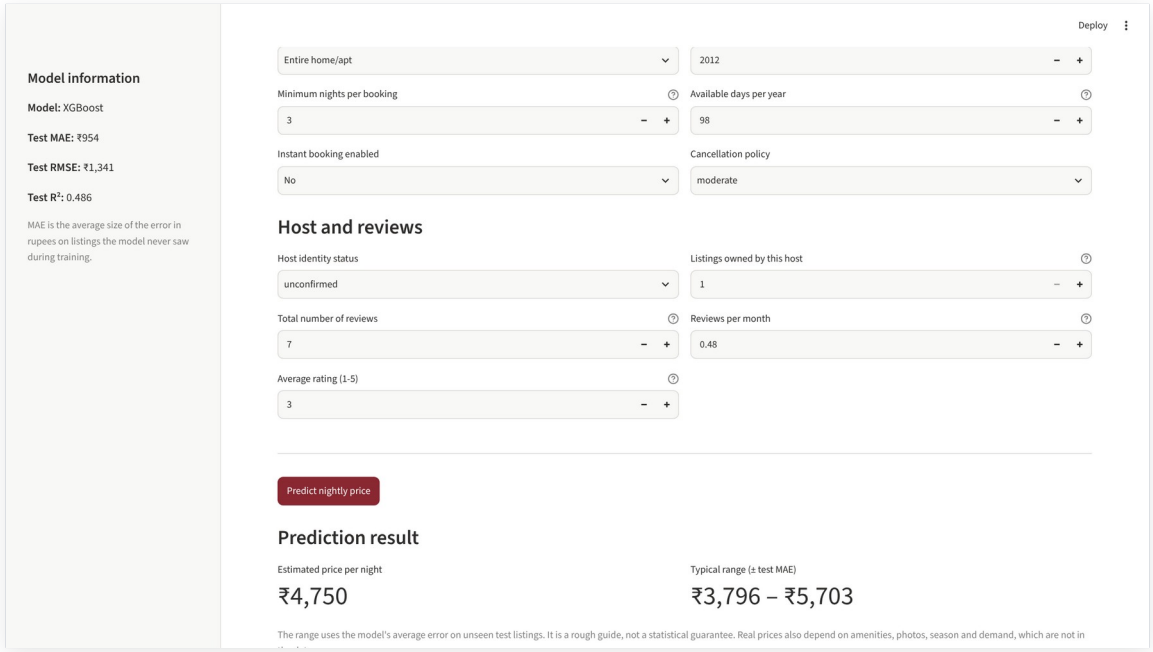
Action: Offer monthly-stay pricing

Dashboard and Price Prediction App



Analytics dashboard (port 8501)

Filter by city, room type and price; see KPIs, price charts, neighbourhoods, reviews, segments and model metrics.



Price prediction app (port 8502)

Enter city, neighbourhood, room type, booking rules and reviews; get a price with a ± MAE range.

1. Host enters listing details
2. Inputs checked against the saved schema
3. Saved XGBoost pipeline predicts
4. Shows ₹4,750 ± ₹954 (default listing)

Verified: real screenshots, 13/13 app tests passed. Not yet deployed online.

Results, Business Insights and Limitations

2.0×
entire home vs private room price

₹3,450 vs ₹2,200
Mumbai vs Jaipur median

₹954
average prediction error (R^2 0.486)

39%
listings look dormant

Airbnb hosts

- Benchmark against the same room type in the same neighbourhood
- Use the app's estimate as a starting price
- Open but rarely booked? Review price and photos

Guests

- Private rooms cost about half of entire homes
- Neighbourhoods in one city differ up to 1.56×
- Higher ratings did not cost more in this data

Market analysts

- Room type and location explain most price variation
- Segment by activity: 39% dormant, 21% under-booked
- Report effect sizes, not just p-values

Honest limitations

- Data quality: source undocumented, possibly synthetic
- Missing values: small; last_review 15.5% empty
- No amenities, size, photos or season
- Model error: ~₹954 average, luxury under-priced
- Generalisation: tested only within this dataset

Conclusion and Future Scope

WHAT WAS ACHIEVED

- Clean, documented analysis of 102,599 listings
- Room type and location proven as main price drivers
- XGBoost model ~30% better than baseline
- Four actionable market segments
- Two tested Streamlit apps

FUTURE SCOPE (PROPOSED)

01 More recent, documented listing data

02 Amenities, size, photos and seasonality

03 Advanced models and explainable AI (e.g. SHAP)

04 More cities and regions across India

05 Live data integration and model refresh

06 Online deployment on Streamlit Cloud

Thank You

Questions & Answers

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